2. Fundamentals of Data Science and Big Data Analytics

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2. Types of Data
3. Tools and Techniques
4. Data Collection and Storage
5. Standard vs Big Data Analytics

Introduction to Data Science

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- This information doesn't come neatly organized and ready to work with.
- Therefore, the purpose of data science is to work with some of that data and drive better-informed decision-making.

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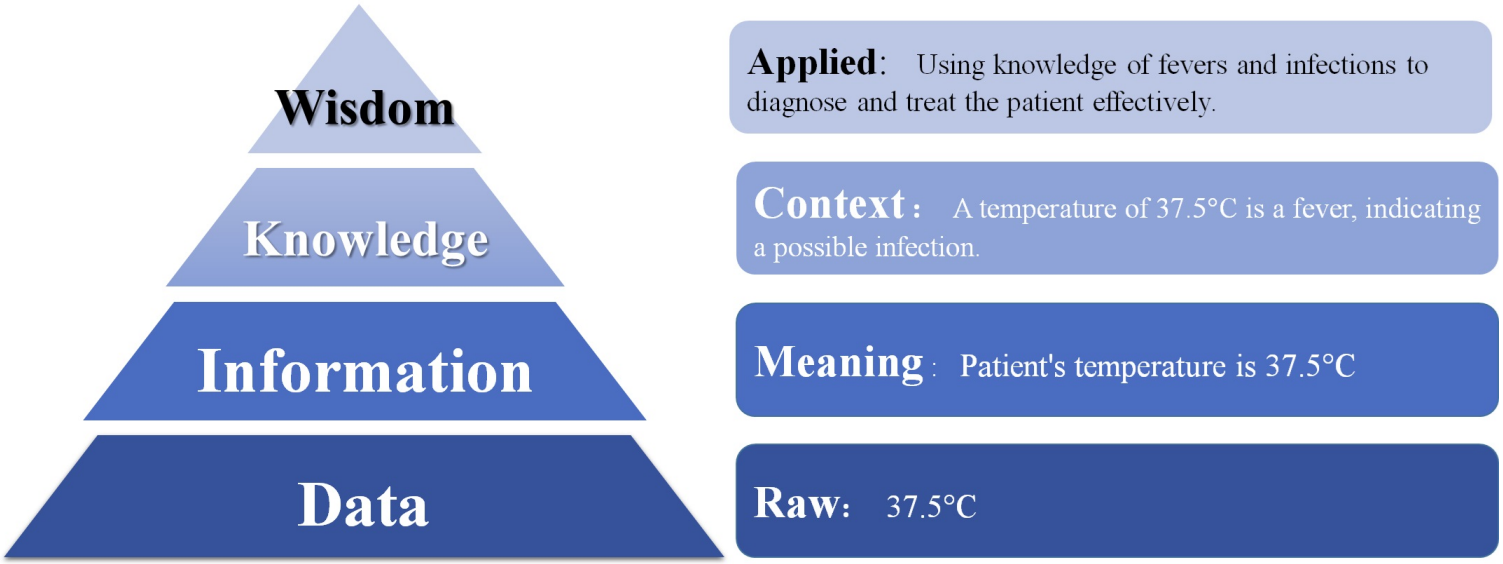
- Build clarity so sales, marketing, and finance all speak the same language
- Spot opportunities before they become challenges

Introduction to Data Science



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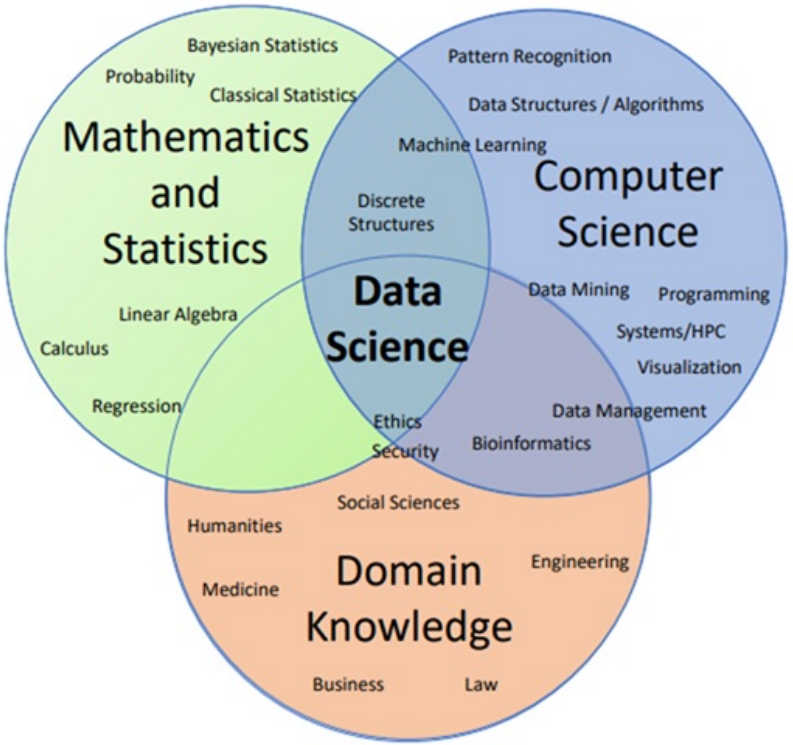

- Data science is all of these things, but it is more than the sum of its parts because of the applications.

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- Challenge - Real data often do not follow regular patterns/ match standard equations

Types of Data

- Structured data: tructured data is highly organized and formatted in a predefined schema, typically in rows and columns
  - Organized in rows and columns (tables)
  - Easy to store and analyze
  - Fast processing and analysis
  - For example: Traffic counts (vehicles per hour), Bus schedules

time	location	vehicle_count	speed
08:00	Bole	230	35
09:00	Bole	320	20

Types of Data

- Semi-structured: Semi-structured data does not follow a strict table format but contains tags, keys, or markers to organize data.
  - Not strictly tabular but has some organization (tags, keys)
  - Hierarchical or nested structure
  - Requires parsing before analysis
  - Example: GPS logs, sensor data

```
{
  "location": "Bole",
  "traffic": {
    "cars": 200,
    "buses": 30
  }
}
```

Types of Data

- Unstructured data: Unstructured data has no predefined format or organization, making it the most complex type
  - No predefined format
  - Difficult to analyze directly
  - Example:- Image, video

Types of Data

Feature	Structured	Semi-Structured	Unstructured
Format	Fixed table	Flexible (tags/keys)	No format
Schema	Strict	Partial	None
Ease of Analysis	Easy	Moderate	Difficult
Storage	Relational DB	NoSQL / files	Data lakes
Examples	Tables, SQL DB	JSON, XML	Text, images, video

Data Type	Example
Structured	Traffic counts, bus schedules
Semi-Structured	GPS logs, sensor data
Unstructured	CCTV videos, social media complaints

Types of Data

Qualitative vs Quantitative Data

- Qualitative data describes qualities or characteristics and is non-numeric.
- Quantitative data represents numerical values that can be measured and analyzed mathematically.

Aspect	Qualitative Data	Quantitative Data
Nature	Descriptive	Numerical
Format	Categories	Numbers
Analysis	Classification	Mathematical/statistical
Examples	Vehicle type, road type	Speed, traffic volume

Tools and Techniques

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   - Disadvantages: May not fit your exact problem

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- Manual collection: Traffic surveys, Road observations, Passenger interviews

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Example Workflow Scenario: Traffic Analysis in Addis Ababa

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4. Analyze congestion patterns

Standard vs Big Data

- Standard Data Science: Focuses on extracting insights and building models from structured or moderately sized datasets using statistical and machine learning techniques.

Unit of Data size	Exact size	Approximate Size	Examples
KB (kilobyte)	2^{10} or 1024 bytes	(10^3 or one thousand) bytes	A typical joke =1KB
MB(megabyte)	2^{20} bytes	(10^6 or one million) bytes	Complete work of Shakespeare =5MB
GB (gigabyte)	2^{30} bytes	(10^9 or one billion) bytes	Ten yards of books on a shelf = 1GB
TB (terabyte)	2^{40} bytes	(10^{12} or one trillion) bytes	All the X-rays for a large hospital =1TB; Tweets created daily =121TB;
PB (peta byte)	2^{50} bytes	(10^{15} or one quadrillion) bytes	All U.S. academic research libraries = 2PB: Data processed in a day by Google =24PB
EB (exa byte)	2^{60} bytes	(10^{18} or one Quintillion) bytes	Total global data created in 2006 = 161EB
ZB (zetta byte)	2^{70} bytes	(10^{21} or one Sextillion) bytes	Total amount of global data created in 2012 = 2.7 ZB and 149 ZB by 2024

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- Standard Data Science: Focuses on extracting insights and building models from structured or moderately sized datasets using statistical and machine learning techniques.
- Big Data: Refers to handling and analyzing very large, complex, and fast-growing datasets that cannot be processed using traditional tools.

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- Value – extracting useful insights

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Tools for Big Data Analytis

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Application examples

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Discussion